Vector Spaces I–II: Subspaces, Span, Independence, Basis and Dimension

Problem sheet for the Day 3 classes · CSIR NET / GATE previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. GATE 2016 MCQ Let $\{x, y, z\}$ be a basis of $\mathbb{R}^3$. Consider the statements P: $\{x+y,\ y+z,\ x-z\}$ is a basis of $\mathbb{R}^3$; Q: $\{x+y+z,\ x+2y-z,\ x-3z\}$ is a basis of $\mathbb{R}^3$. Which of the following holds?

(A) both P and Q
(B) only P
(C) only Q
(D) neither P nor Q

2. CSIR NET Dec 2019 MCQ Let $W = \{A = (a_{ij}) \in M_4(\mathbb{R}) : \sum_{i+j=k} a_{ij} = 0 \text{ for } k = 2, 3, \ldots, 8\}$. Then $\dim W$ is

(A) 7
(B) 8
(C) 9
(D) 10

3. CSIR NET Dec 2016 MCQ Which of the following sets of functions from $\mathbb{R}$ to $\mathbb{R}$ is a vector space over $\mathbb{R}$? $S_1 = \{f \mid \lim_{x\to 3} f(x) = 0\}$, $S_2 = \{g \mid \lim_{x\to 3} g(x) = 1\}$, $S_3 = \{h \mid \lim_{x\to 3} h(x) \text{ exists}\}$.

(A) Only S_1
(B) Only S_2
(C) S_1 and S_3 but not S_2
(D) All the three are vector spaces

4. CSIR NET June 2016 MCQ Let $x = (x_1, x_2, x_3),\ y = (y_1, y_2, y_3) \in \mathbb{R}^3$ be linearly independent. Let $\delta_1 = x_2y_3 - y_2x_3,\ \delta_2 = x_1y_3 - y_1x_3,\ \delta_3 = x_1y_2 - y_1x_2$. If V is the span of x, y, then

(A) $V = \{(u, v, w) : \delta_1 u - \delta_2 v + \delta_3 w = 0\}$
(B) $V = \{(u, v, w) : -\delta_1 u + \delta_2 v + \delta_3 w = 0\}$
(C) $V = \{(u, v, w) : \delta_1 u + \delta_2 v - \delta_3 w = 0\}$
(D) $V = \{(u, v, w) : \delta_1 u + \delta_2 v + \delta_3 w = 0\}$

5. CSIR NET June 2017 MCQ Let A be a 4×4 matrix. Suppose that the null space $N(A)$ of A is $\{(x, y, z, w) \in \mathbb{R}^4 : x + y + z = 0,\ x + y + w = 0\}$. Then

(A) $\dim(\text{column space}(A)) = 1$
(B) $\dim(\text{column space}(A)) = 2$
(C) $\text{rank}(A) = 1$
(D) $S = \{(1, 1, 1, 0), (1, 1, 0, 1)\}$ is a basis of $N(A)$

6. CSIR NET June 2017 MCQ Let $p(x) = \alpha x^2 + \beta x + \gamma$ be a polynomial, where $\alpha, \beta, \gamma \in \mathbb{R}$. Fix $x_0 \in \mathbb{R}$. Let $S = \{(a, b, c) \in \mathbb{R}^3 : p(x) = a(x - x_0)^2 + b(x - x_0) + c \text{ for all } x \in \mathbb{R}\}$. Then the number of elements in S is

(A) 0
(B) 1
(C) strictly greater than 1 but finite
(D) infinite

7. CSIR NET Dec 2014 MCQ Which of the following matrices has the same row space as the matrix $\begin{pmatrix} 4 & 8 & 4 \\ 3 & 6 & 1 \\ 2 & 4 & 0 \end{pmatrix}$?

(A) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
(B) $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
(C) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
(D) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$

8. CSIR NET June 2019 MCQ Let V denote the vector space of real valued continuous functions on the closed interval $[0, 1]$. Let W be the subspace of V spanned by $\{\sin x, \cos x, \tan x\}$. Then the dimension of W over $\mathbb{R}$ is

(A) 1
(B) 2
(C) 3
(D) infinite

9. CSIR NET Dec 2016 MSQ Let M_n denote the vector space of all $n \times n$ real matrices. Among the following subsets of M_n, decide which are linear subspaces.

(A) $V_1 = \{A \in M_n : A \text{ is nonsingular}\}$
(B) $V_2 = \{A \in M_n : \det(A) = 0\}$
(C) $V_3 = \{A \in M_n : \text{trace}(A) = 0\}$
(D) $V_4 = \{BA : A \in M_n\}$, where B is some fixed matrix in M_n

10. CSIR NET June 2019 MSQ Let $p(x) = a_0 + a_1 x + \cdots + a_n x^n$ be a non-constant polynomial of degree $n \geq 1$. Consider the polynomial $q(x) = \int_0^x p(t)\, dt$, $r(x) = \frac{d}{dx} p(x)$. Let V denote the real vector space of all polynomials in x. Then which of the following are true?

(A) q and r are linearly independent in V
(B) q and r are linearly dependent in V
(C) x^n belongs to the linear span of q and r
(D) x^{n+1} belongs to the linear span of q and r